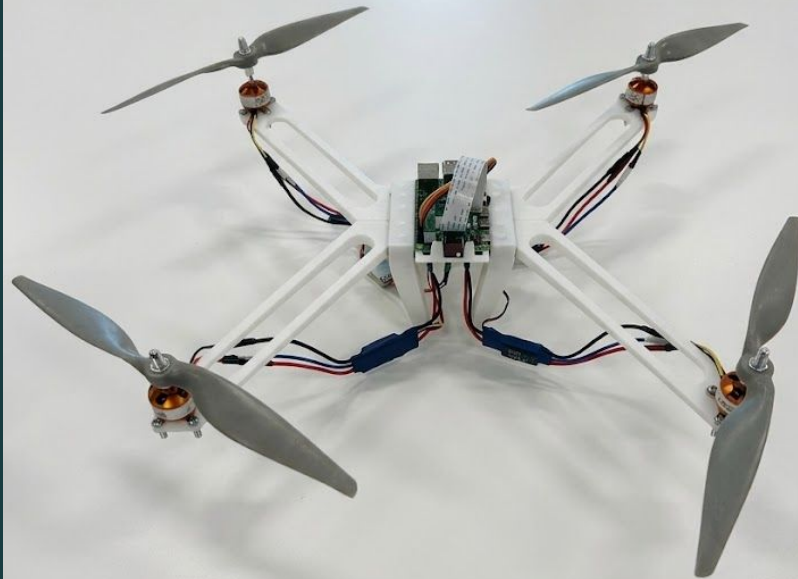


AI Drone Project

Adam DeCristofaro
6/1/26



Content overview

Industry Standard

- Traditionally serve as flying cameras
- More recently AI has been applied to after flight

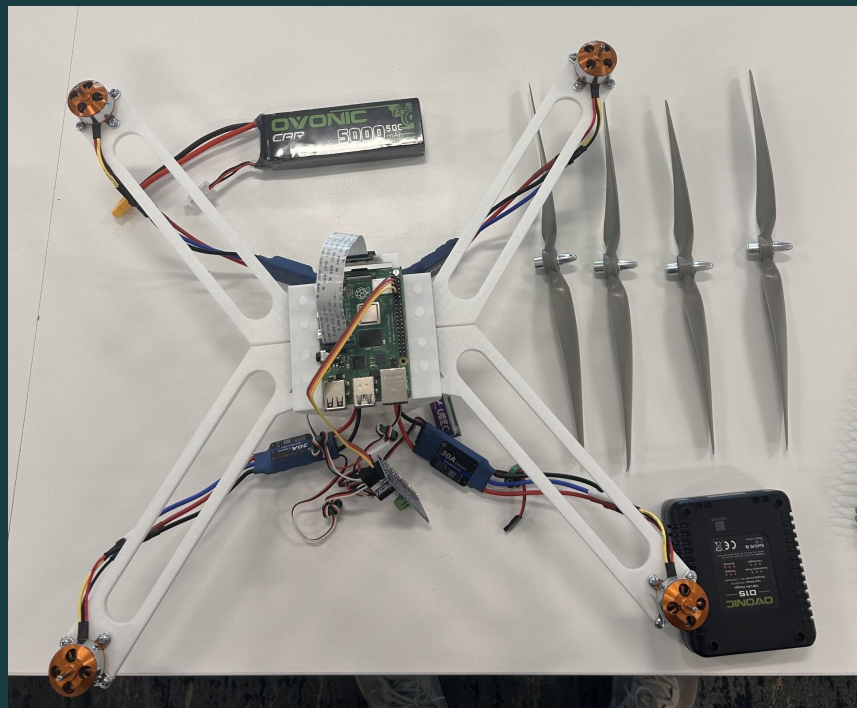
Project Goal

- Create a drone “coworker” who you can communicate with in real time



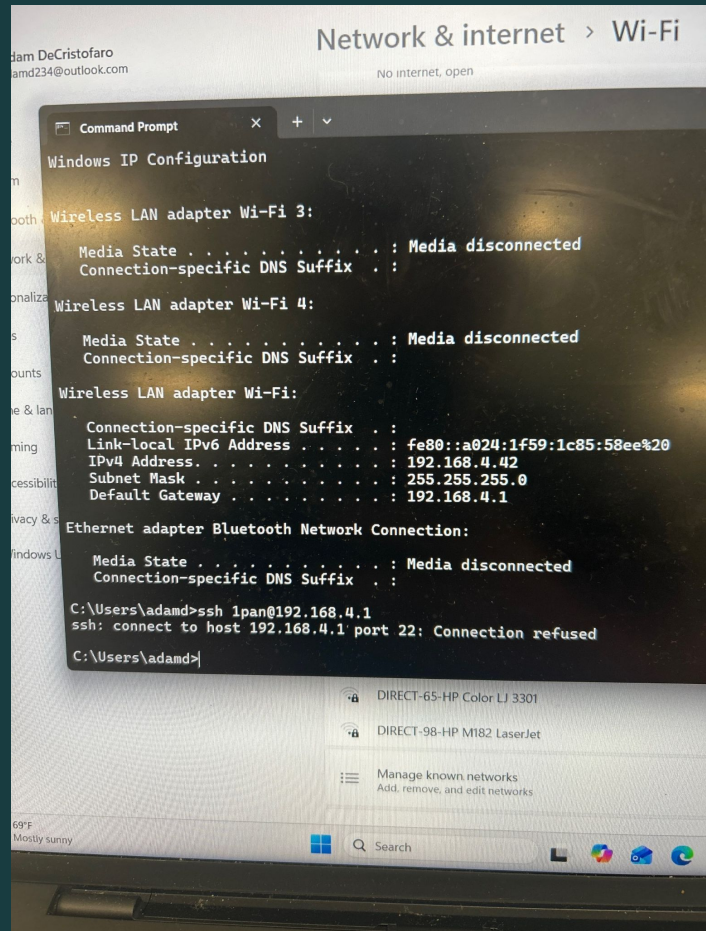
Day 1 Hardware Audit

- All of the parts in the report from last summer are accounted for
- MPU-6050 Gyroscope is currently unmounted
- Drone currently has counterclockwise blades and needs two clockwise blades for yaw balance
- Current blades are too long for the motors size(2600kV)



Day 1 Software Audit

- Successfully connected to the Raspberry Pi WiFi "PiCam"
- Was unable to connect wirelessly as the ssh port was refusing connections
- Plan to access it locally using a monitor to flip the ssh allowing wireless connections



Next Steps

Summer Outline

- Order different blades or motors
- Create an attachment for the gyroscope
- Connect to the raspberry pi to get live footage from the camera
- Test the drone for flight
- Begin developing the AI models for real time detection

Goals for Next Week

- Order the correct blades
- Connect to raspberry pi

Milestones

1. Weeks 1 & 2

Successfully equip the drone with all of the parts necessary for flight

2. Weeks 3-6

Achieve flight with the drone with programmed flight and the use of a controller

Weeks 7-10

Install AI models into the drones software so it can detect defects like a crack in a wall or window and send a message back to the user in real time

End Goal for the Summer

Be able to manually drive the drone inside of Debaun gym and have it detect staged "defects" in the walls and communicate it back to the user